gpeyt1-19 · Pseudostatic rock slope

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A highway rock cut has a daylighting planar joint dipping α = 35° with length L = 60 ft along the joint per ft of cut. The sliding block weighs W = 180,000 lb/ft. Joint strength: c = 500 psf, φ = 30°. The slope is fully drained (no water in the tension crack or on the joint). For seismic screening use k = 0.15. The rock unit weight is 160 pcf and RQD is 72% (not needed).

Most nearly, the pseudostatic factor of safety is:

A. 1.12
B. 0.85
C. 0.92
D. 1.18

Given equations / data

> FS = [cL + (W cos α − kW sin α − PWR − PWJ sin α) tan φ]/(W sin α + kW cos α + PWJ cos α); PWR = PWJ = 0 when drained

Worked solution

1. cL = 500×60 = 30,000 lb/ft
2. Normal: W cos α − kW sin α = 147,447 − 15,487 = 131,961 lb/ft
3. Resisting = 30,000 + 131,961×tan30 = 106,188
4. Driving = W sin α + kW cos α = 125,361
5. FS = 0.85 (static 1.12)

Final answer

0.85

Why the other values are wrong

- **1.12** — static FS (dropped k)
- **0.92** — added k only to the driving side
- **1.18** — inverted FS

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.3 Rock Slope Failure (Stability of Rock Slope figure), p. 104

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=90zXmI_yn_g (no creator values copied).

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